

SOUND

1. Sound is a form of

- A) Matter
- B) Energy
- C) Force
- D) Light

Answer: B

2. Sound is produced due to

- A) Reflection
- B) Vibrations
- C) Refraction
- D) Rotation

Answer: B

3. Sound travels through

- A) Vacuum
- B) Medium
- C) Space only
- D) Light

Answer: B

4. Sound cannot travel through

- A) Solids
- B) Liquids
- C) Gases
- D) Vacuum

Answer: D

5. Sound waves in air are

- A) Transverse waves
- B) Longitudinal waves
- C) Surface waves
- D) Water waves

Answer: B

6. In longitudinal waves particles vibrate

- A) Perpendicular to direction of wave
- B) Parallel to direction of wave
- C) Randomly
- D) In circles

Answer: B

7. A region of high pressure in a sound wave is called

- A) Rarefaction
- B) Compression
- C) Reflection
- D) Diffusion

Answer: B

8. A region of low pressure in a sound wave is called

- A) Compression
- B) Rarefaction
- C) Diffraction
- D) Resonance

Answer: B

9. Number of vibrations per second is called

- A) Amplitude
- B) Wavelength
- C) Frequency
- D) Intensity

Answer: C

10. SI unit of frequency is

- A) Watt
- B) Hertz
- C) Newton
- D) Joule

Answer: B

11. One hertz means

- A) 1 vibration per minute

- B) 1 vibration per second
- C) 10 vibrations per second
- D) 100 vibrations per second

Answer: B

12. The loudness of sound depends on

- A) Frequency
- B) Amplitude
- C) Speed
- D) Density

Answer: B

13. Pitch of sound depends on

- A) Amplitude
- B) Frequency
- C) Speed
- D) Medium

Answer: B

14. High pitch sound has

- A) Low frequency
- B) High frequency
- C) Low amplitude
- D) No vibration

Answer: B

15. Low pitch sound has

- A) High frequency
- B) Low frequency
- C) High amplitude
- D) Zero amplitude

Answer: B

16. Speed of sound in air is about

- A) 100 m/s
- B) 330 m/s
- C) 500 m/s

D) 1000 m/s

Answer: B

17. Speed of sound is maximum in

A) Solids

B) Liquids

C) Gases

D) Vacuum

Answer: A

18. Speed of sound is minimum in

A) Solids

B) Liquids

C) Gases

D) Metals

Answer: C

19. Sound travels faster in

A) Air

B) Water

C) Vacuum

D) Gas

Answer: B

20. Audible sound range for humans is

A) 20–200 Hz

B) 20–2000 Hz

C) 20–20000 Hz

D) 200–20000 Hz

Answer: C

21. Sound below 20 Hz is called

A) Ultrasound

B) Infrasound

C) Audible sound

D) Echo

Answer: B

22. Sound above 20000 Hz is called

- A) Ultrasound
- B) Infrasound
- C) Audible sound
- D) Noise

Answer: A

23. Bats use

- A) Ultrasound
- B) Infrasound
- C) Audible sound
- D) Light

Answer: A

24. Dolphins communicate using

- A) Ultrasound
- B) Infrasound
- C) Light
- D) Heat

Answer: A

25. Reflection of sound is called

- A) Echo
- B) Refraction
- C) Diffraction
- D) Absorption

Answer: A

26. Echo is produced due to

- A) Reflection of sound
- B) Absorption
- C) Refraction
- D) Diffusion

Answer: A

27. Echo is heard when reflected sound reaches after

- A) 0.1 s

B) 0.5 s

C) 1 s

D) 2 s

Answer: A

28. Minimum distance for echo in air is about

A) 10 m

B) 17 m

C) 34 m

D) 50 m

Answer: C

29. SONAR stands for

A) Sound Navigation and Ranging

B) Sound Navigation and Recording

C) Sound Number and Range

D) Sound Navigation Radar

Answer: A

30. SONAR is used to

A) Measure distance underwater

B) Measure temperature

C) Measure pressure

D) Measure density

Answer: A

31. SONAR uses

A) Infrared waves

B) Ultrasonic waves

C) Radio waves

D) Light waves

Answer: B

32. Ultrasound is used in

A) Medical imaging

B) Lighting

C) Cooling

D) Painting

Answer: A

33. Ultrasound can detect

A) Cracks in metals

B) Colour of metals

C) Density only

D) Heat

Answer: A

34. Ultrasonic waves are used to

A) Break kidney stones

B) Melt metals

C) Produce light

D) Generate electricity

Answer: A

35. Instrument used to hear heart beat is

A) Telescope

B) Microscope

C) Stethoscope

D) Barometer

Answer: C

36. Stethoscope works on principle of

A) Reflection of sound

B) Absorption

C) Refraction

D) Diffusion

Answer: A

37. Wavelength is distance between

A) Two compressions

B) Two rarefactions

C) Two successive points

D) All of these

Answer: D

38. Unit of wavelength is

- A) Meter
- B) Hertz
- C) Newton
- D) Joule

Answer: A

39. Time taken for one vibration is called

- A) Frequency
- B) Period
- C) Pitch
- D) Loudness

Answer: B

40. Unit of time period is

- A) Second
- B) Meter
- C) Hertz
- D) Watt

Answer: A

41. Frequency and time period are

- A) Directly proportional
- B) Inversely proportional
- C) Equal
- D) Independent

Answer: B

42. Sound intensity depends on

- A) Amplitude
- B) Frequency
- C) Speed
- D) Density

Answer: A

43. Noise is

- A) Pleasant sound

- B) Unwanted sound
- C) Musical sound
- D) High frequency sound

Answer: B

44. Noise pollution affects

- A) Hearing
- B) Vision
- C) Taste
- D) Smell

Answer: A

45. Musical sound is

- A) Pleasant
- B) Unpleasant
- C) Loud
- D) Sharp

Answer: A

46. Tuning fork produces

- A) Light
- B) Sound
- C) Heat
- D) Electricity

Answer: B

47. When tuning fork vibrates it produces

- A) Heat
- B) Sound waves
- C) Light waves
- D) Electric waves

Answer: B

48. Thunder is heard after lightning because

- A) Sound travels slower than light
- B) Sound travels faster
- C) Light travels slower

D) Sound stops

Answer: A

49. Sound travels faster when temperature

A) Decreases

B) Increases

C) Stops

D) Remains constant

Answer: B

50. Intensity of sound depends on

A) Distance from source

B) Amplitude

C) Medium

D) All of these

Answer: D

51. Sound wave transfers

A) Matter

B) Energy

C) Mass

D) Weight

Answer: B

52. Sound travels through solids by

A) Particle vibration

B) Light

C) Electricity

D) Heat

Answer: A

53. Humans cannot hear

A) Ultrasound

B) Audible sound

C) Musical sound

D) Normal sound

Answer: A

54. Dogs can hear

- A) Ultrasound
- B) Only audible sound
- C) Only infrasound
- D) No sound

Answer: A

55. Whale communication uses

- A) Infrasound
- B) Ultrasound
- C) Light
- D) Heat

Answer: A

56. Loudness increases with

- A) Frequency
- B) Amplitude
- C) Speed
- D) Wavelength

Answer: B

57. Pitch increases with

- A) Frequency
- B) Amplitude
- C) Wavelength
- D) Speed

Answer: A

58. Sound waves are

- A) Mechanical waves
- B) Light waves
- C) Electromagnetic waves
- D) Magnetic waves

Answer: A

59. Mechanical waves require

- A) Vacuum

B) Medium

C) Space

D) Heat

Answer: B

60. Echo is useful in

A) Measuring distance

B) Measuring light

C) Measuring pressure

D) Measuring heat

Answer: A

61. Echo sounding is used to find

A) Depth of sea

B) Height of mountain

C) Distance to sun

D) Air pressure

Answer: A

62. Bats detect obstacles using

A) Echolocation

B) Reflection of light

C) Heat

D) Magnetism

Answer: A

63. Sound waves consist of

A) Compression and rarefaction

B) Light and shadow

C) Heat and cold

D) Waves and rays

Answer: A

64. Amplitude is maximum displacement of

A) Particle

B) Light

C) Heat

D) Pressure

Answer: A

65. Greater amplitude produces

A) Louder sound

B) Lower sound

C) No sound

D) Silence

Answer: A

66. Shorter wavelength means

A) Higher frequency

B) Lower frequency

C) Zero frequency

D) Same frequency

Answer: A

67. Sound cannot travel on

A) Earth

B) Moon

C) Water

D) Air

Answer: B

68. Sound energy is produced by

A) Vibrating bodies

B) Moving light

C) Heat

D) Pressure

Answer: A

69. Human ear detects

A) Sound vibrations

B) Light vibrations

C) Heat vibrations

D) Pressure vibrations

Answer: A

70. Speed of sound depends on

- A) Medium
- B) Temperature
- C) Density
- D) All of these

Answer: D

71. In vacuum sound speed is

- A) High
- B) Low
- C) Zero
- D) Infinite

Answer: C

72. Ultrasonic waves are used in

- A) SONAR
- B) Radio
- C) Television
- D) Radar

Answer: A

73. Reflection of sound follows

- A) Laws of reflection
- B) Laws of motion
- C) Laws of gravity
- D) Laws of pressure

Answer: A

74. Hard surfaces

- A) Reflect sound
- B) Absorb sound
- C) Destroy sound
- D) Produce sound

Answer: A

75. Soft surfaces

- A) Reflect sound

- B) Absorb sound
- C) Produce sound
- D) Increase sound

Answer: B

76. Curtains reduce

- A) Sound reflection
- B) Light reflection
- C) Heat reflection
- D) Pressure

Answer: A

77. Carpets reduce

- A) Noise
- B) Light
- C) Heat
- D) Pressure

Answer: A

78. Auditorium walls are covered with

- A) Soft materials
- B) Metals
- C) Glass
- D) Plastic

Answer: A

79. Sound waves carry

- A) Energy
- B) Mass
- C) Volume
- D) Density

Answer: A

80. Vibrations produce

- A) Sound
- B) Light
- C) Heat

D) Electricity

Answer: A

81. High frequency means

A) More vibrations

B) Fewer vibrations

C) No vibration

D) Same vibration

Answer: A

82. Sound spreads in

A) All directions

B) One direction

C) Two directions

D) No direction

Answer: A

83. Sound needs

A) Particles

B) Light

C) Electricity

D) Magnetism

Answer: A

84. Sound energy converts to

A) Hearing sensation

B) Vision

C) Taste

D) Smell

Answer: A

85. Human ear sensitivity is about

A) 20 Hz

B) 200 Hz

C) 2000 Hz

D) 20000 Hz

Answer: A

86. High frequency sound is

- A) Shrill
- B) Dull
- C) Silent
- D) Heavy

Answer: A

87. Low frequency sound is

- A) Deep
- B) Shrill
- C) Loud
- D) Sharp

Answer: A

88. Sound reflection is used in

- A) Megaphones
- B) Lamps
- C) Fans
- D) Motors

Answer: A

89. Megaphone increases

- A) Loudness
- B) Light
- C) Heat
- D) Pressure

Answer: A

90. Speed of sound in solids is

- A) Highest
- B) Lowest
- C) Same as air
- D) Zero

Answer: A

91. Sound energy decreases with

- A) Distance

- B) Frequency
- C) Amplitude
- D) Time

Answer: A

92. Two whales communicate because sound travels faster in

- A) Water
- B) Air
- C) Vacuum
- D) Space

Answer: A

93. Vibrations of drum produce

- A) Sound
- B) Light
- C) Heat
- D) Electricity

Answer: A

94. Human voice is produced by

- A) Vocal cords
- B) Lungs
- C) Nose
- D) Teeth

Answer: A

95. Vibrating strings produce

- A) Musical sound
- B) Light
- C) Heat
- D) Pressure

Answer: A

96. Speed of sound increases with

- A) Temperature
- B) Pressure
- C) Colour

D) Shape

Answer: A

97. Reflection of sound requires

A) Surface

B) Light

C) Heat

D) Vacuum

Answer: A

98. Sound travels slower than

A) Light

B) Air

C) Water

D) Metal

Answer: A

99. Sound energy is detected by

A) Ear

B) Eye

C) Nose

D) Skin

Answer: A

100. Basic requirement for sound propagation is

A) Medium

B) Vacuum

C) Light

D) Electricity

Answer: A